Computer Networks 1B Revision Questions

CONE5112 and CONE5112w | All six learning units | Questions only

These questions include short questions, long questions and scenario-based questions. No multiple-choice or true-and-false questions are included.

Instructions to students

Answer short questions briefly, using examples where requested. Answer long questions in structured paragraphs, explaining how and why the concepts work. Apply your answers to the facts given in each scenario. Show formulas, calculations and units for numerical questions. Use labelled diagrams where they help explain your answer.

Learning Unit 1 Computer Networks and Wireless Mobility

Short questions

1.1 Explain what is meant by describing the Internet as a network of networks. Identify the roles of hosts, routers and Internet service providers.

1.2 Distinguish between packet switching and circuit switching. Give one practical example of each.

1.3 Explain the difference between transmission delay and propagation delay.

1.4 Define throughput and explain why a connection's actual throughput may be lower than its advertised bandwidth.

1.5 Explain encapsulation using an example of a student sending a message over a network.

Long questions

1.6 Explain the responsibilities of the five layers in the Internet teaching model. Trace how application data moves down the sender's protocol stack and up the receiver's stack.

1.7 Compare fibre, Ethernet, Wi-Fi and cellular access. Discuss their suitability for a fixed computer laboratory, a lecturer working from home and a student travelling between locations.

1.8 Explain how CDMA separates users sharing a wireless medium. Discuss the purpose of spreading codes and the near-far problem.

1.9 Compare the basic architectures of 4G LTE and standalone 5G. Explain how radio access and core-network components support connectivity and mobility.

1.10 Evaluate Mobile IP as a method of supporting device mobility. Explain home addresses, care-of addresses, home agents and tunnelling, including benefits and limitations.

Scenario questions

A college has 40 learners sharing a Wi-Fi connection during online classes. Its Internet connection operates at 50 Mb/s. Learners experience video delays when several computers download updates. One learner also attends classes using a mobile phone while travelling between cellular coverage areas.

A 1,500-byte packet crosses a 10 Mb/s link over a distance of 100 km. Assume a propagation speed of 200,000,000 metres per second.

1.11 Analyse why the learners may experience high delay even though the college has a 50 Mb/s Internet connection. Distinguish between possible wireless and Internet-uplink problems.

1.12 Calculate the packet's transmission delay and propagation delay. Explain which delay would change if the link rate doubled.

1.13 Explain how handover supports the travelling learner's connection. Discuss two conditions that could cause a temporary interruption.

Learning Unit 2 The Application Layer

Short questions

2.1 Distinguish between client-server and peer-to-peer application architectures.

2.2 Explain what a socket is and why network applications use port numbers.

2.3 Explain the purpose of DNS. Distinguish between A, AAAA and MX records.

2.4 Explain the difference between a browser cookie and a web cache.

2.5 Describe how DASH adjusts video delivery when available throughput changes.

Long questions

2.6 Explain the sequence of events when a student enters the name of a learning portal and requests a page using HTTPS over TCP. Include DNS resolution, transport connection establishment, security and HTTP communication.

2.7 Compare TCP and UDP from an application developer's perspective. Explain their suitability for assignment uploads, live audio and short request-response exchanges.

2.8 Explain how persistent connections, caching and HTTP/2 can improve web performance. Discuss limitations and privacy considerations.

2.9 Explain how a CDN supports multimedia delivery. Discuss server placement, request distribution, caching and the relationship between a CDN and adaptive streaming.

2.10 Describe the main steps in TCP socket programming for a client and server. Explain why applications must define message boundaries even when TCP provides reliable delivery.

Scenario questions

A college hosts its learning portal on a server with a 100 Mb/s upload connection. At the start of a lecture, 100 students request the same 100 MB video file. The portal also stores login sessions and displays individual student marks. Some students use unstable mobile connections.

2.11 Calculate the ideal minimum time required for the server to upload a separate copy of the file to all 100 students. Assume decimal units, no overhead and no other bottlenecks.

2.12 Recommend how caching, a CDN and DASH could improve the students' experience. Explain the different contribution of each mechanism.

2.13 Explain how the portal should treat public video content differently from private marks when applying caching and session-management mechanisms.

Learning Unit 3 The Transport Layer

Short questions

3.1 Explain how transport-layer communication differs from network-layer communication.

3.2 Distinguish between transport-layer multiplexing and demultiplexing.

3.3 Identify the four fields in a UDP header and explain the purpose of each.

3.4 Explain what a TCP sequence number represents and what an acknowledgement number indicates.

3.5 Distinguish between TCP flow control and congestion control.

Long questions

3.6 Explain how sequence numbers, acknowledgements, timers and retransmissions support reliable data transfer over an unreliable network.

3.7 Compare stop-and-wait, Go-Back-N and Selective Repeat. Discuss efficiency, receiver behaviour and recovery after packet loss.

3.8 Explain TCP's three-way handshake and how TCP provides reliable, ordered delivery after the connection is established.

3.9 Analyse the causes and costs of network congestion. Explain classic TCP slow start, congestion avoidance, fast retransmit and fast recovery.

3.10 Discuss fairness when multiple transport flows share a bottleneck. Explain the role of ECN and how protocols such as QUIC extend transport design.

Scenario questions

A server sends assignment files to a student's laptop. The laptop has limited receive-buffer space. At the same time, the campus uplink is congested by large downloads.

One TCP data segment starts at sequence number 4,000 and carries 800 bytes. Assume that all earlier bytes have arrived. In a separate reliable-transfer exercise, packets 1, 2, 3, 4 and 5 are transmitted, but packet 3 is lost.

3.11 Determine the expected TCP acknowledgement number after the 800-byte segment arrives. Explain your calculation.

3.12 Explain how the receiver window and congestion window could independently restrict the server's sending behaviour.

3.13 Compare how basic Go-Back-N and Selective Repeat would recover from the loss of packet 3. State your assumptions about receiver behaviour.

Learning Unit 4 The Network Layer Data Plane

Short questions

4.1 Distinguish between routing and forwarding.

4.2 Explain the purpose of a router's input ports, switching fabric and output ports.

4.3 Explain longest-prefix matching and the role of a default route.

4.4 Define head-of-line blocking and explain where it can occur.

4.5 Distinguish between IPv4 and IPv6 addressing, and identify two differences between their basic packet headers.

Long questions

4.6 Compare traditional networking and SDN. Discuss decision-making, programmability, policy consistency and failure handling.

4.7 Explain the Internet's best-effort service model. Evaluate its suitability for file transfer, live video and delay-sensitive control applications.

4.8 Compare FIFO, priority, round-robin and weighted scheduling. Discuss how their use affects delay and resource sharing.

4.9 Explain how to plan IPv4 subnets for departments of different sizes. Include host requirements, prefix selection, address ranges and non-overlapping allocation.

4.10 Explain generalised forwarding and match-action rules. Discuss OpenFlow and the roles of firewalls, NAT devices and load balancers as middleboxes.

Scenario questions

A college has the network block 192.168.40.0/24. It needs separate subnets for a student laboratory with 50 computers and one gateway, and an administration office with 20 computers and one gateway.

A router contains the following forwarding entries.

Destination prefix	Outgoing interface
10.0.0.0/8	A
10.20.0.0/16	B
10.20.5.0/24	C
0.0.0.0/0	D

Live classes and backups compete for a 12 Mb/s uplink.

4.11 Allocate suitable non-overlapping subnets to the laboratory and administration office. Show each prefix, subnet mask, network address, usable range and broadcast address.

4.12 Determine the outgoing interface for packets addressed to 10.20.5.18, 10.20.8.9 and 172.16.1.10. Explain each choice.

4.13 Recommend a scheduling approach for the shared uplink. Explain why increasing buffer memory alone may not solve live-class delays.

Learning Unit 5 The Network Layer Control Plane

Short questions

5.1 Explain the difference between the data plane, control plane and network management.

5.2 Define an autonomous system and distinguish between intra-domain and inter-domain routing.

5.3 Explain the main difference between link-state and distance-vector routing.

5.4 Distinguish between unicast, multicast and anycast.

5.5 Explain what ping and traceroute can reveal, and identify one limitation of each.

Long questions

5.6 Compare OSPF and BGP. Discuss their operating scope, exchanged information and route-selection principles.

5.7 Explain how link-state and distance-vector algorithms calculate routes. Discuss convergence, overhead and routing-loop risks.

5.8 Analyse how business relationships and administrative policies influence BGP route selection and advertisement.

5.9 Explain how SNMP, NETCONF and YANG support network management. Distinguish monitoring, configuration operations and data modelling.

5.10 Evaluate anycast for a distributed DNS service. Discuss routing-based selection, resilience and challenges for stateful applications.

Scenario questions

A college connects buildings A, B and C with these link costs.

Link	Cost
A to B	3
A to C	10
B to C	2

The college uses OSPF internally and connects to two Internet providers. A network administrator observes that a remote website loads successfully, although the server does not respond to ping. An interface counter increases by 25,000,000 bytes over 10 seconds.

5.11 Calculate the lower-cost path from A to C. Explain how routing information and forwarding behaviour would change if the A-to-B link failed.

5.12 Explain why the college's preferred Internet route might not have the shortest AS path. Also explain why failed ping does not prove that the remote server is offline.

5.13 Calculate the interface's average traffic rate in Mb/s. Explain how monitoring this rate over time could support capacity planning.

Learning Unit 6 The Link Layer and LANs

Short questions

6.1 Explain the main functions of the link layer and identify which services vary between technologies.

6.2 Distinguish between a MAC address, an IP address and a transport port number.

6.3 Explain the purpose of ARP and when a host resolves its gateway's MAC address.

6.4 Distinguish between error detection and error correction.

6.5 Explain the purpose of DHCP and describe the DHCPv4 DORA sequence.

Long questions

6.6 Explain how frames and IP packets are handled as data crosses switches and routers from a source host to a remote destination.

6.7 Compare parity checking, checksums and CRC. Discuss their operation, capabilities and limitations.

6.8 Compare channel partitioning, random access and taking-turns protocols. Explain how Wi-Fi access differs from modern switched full-duplex Ethernet.

6.9 Explain how an Ethernet switch learns MAC addresses and handles known unicast, unknown unicast and broadcast frames.

6.10 Evaluate VLANs and modern data centre architecture as methods of supporting segmentation, performance, resilience and application hosting.

Scenario questions

A college places students in VLAN 10 and staff in VLAN 20. Students should access the learning server but must not access payroll services.

A student computer receives IP address 192.168.10.25, subnet mask 255.255.255.0 and gateway 192.168.10.1. The student wants to access a remote server at 203.0.113.20. The switch has not yet learned the gateway's MAC address. Another student has a valid IP configuration but cannot resolve website names.

6.11 Explain how the first student identifies the correct next hop, obtains its MAC address and sends the first frame toward the remote server.

6.12 Explain how access ports, trunks, inter-VLAN routing and access controls should support the college's requirements.

6.13 Propose a troubleshooting sequence for the name-resolution failure. Explain how DHCP, DNS, UDP, IP and local link delivery contribute to successful communication.

Integrated Long Scenario

Ubuntu College is expanding its network to support 120 students, 30 staff members and several application servers. It currently uses one flat LAN, a 50 Mb/s Internet uplink and Wi-Fi access points distributed across three buildings.

Students report that live classes freeze during large software downloads; some computers receive incorrect network settings; the learning portal sometimes works by IP address but fails by name; and users experience interruptions while moving between wireless coverage areas. Staff are concerned that students could access confidential systems.

The college wants separate networks, more consistent management and a resilient connection between buildings.

7.1 Design a suitable logical network structure. Explain subnet sizing, VLAN allocation, gateways and controlled access to shared servers.

7.2 Analyse the reported symptoms. For each symptom, identify plausible causes, suitable diagnostic evidence and an appropriate corrective action.

7.3 Trace a student's HTTPS portal request from initial network configuration to receipt of the page. Explain the role of each relevant layer and protocol.

7.4 Recommend how the college should manage congestion and improve multimedia delivery. Discuss scheduling, download policies, caching, CDNs and adaptive streaming.

7.5 Evaluate traditional routing and SDN for the expanded network. Discuss scalability, configuration consistency, resilience and security.

7.6 Develop a network-management approach using monitoring, secure configuration and documented changes. Explain how the college would verify that its improvements have worked.